

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2bi2____ / 2IPbi2__

REGISTRATION NUMBER

BIOLOGY

9648/03

Paper 3

31 August 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

Sections A - C

Answer **all** questions in the spaces provided on the question paper.

Sections D - E

Answer **all** questions on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 14
2	/ 14
3	/ 12
4	/ 12
5	/ 20
TOTAL	/ 72

This document consists of **11** printed pages.

[Turn over

Section A

Answer the question in this section.

- 1** In a maternity ward at a local hospital, a mix-up involving three couples and three babies caused a lot of confusion. Based on phenotypic characteristics, the nurses were unable to correctly identify the parents of the babies. In order to solve the case, a scientist was called in to carry out a DNA test to identify the parents of the babies. The test was based on the principle that different individuals have a different number of repeating units at a particular locus in a chromosome.

Chromosome 13 was isolated from the DNA samples that were obtained from the three couples and three babies and used for further analysis. The sequence below shows a segment of chromosome 13, which was used in the analysis where (TTAGGAT) is the repeating unit and n is the number of repeats.

5' ...GCTAAGTATTGCTCAAGA... (TTAGGAT)n...GATAAATAACTGGCTAGTA...-3'
 3' ...CGATTCATAACGAGTTCT... (AATCCTA)n... CTATTTATTGACCGATCAT...-5'

Fig. 1.1 shows the results of the DNA test obtained from each individual.

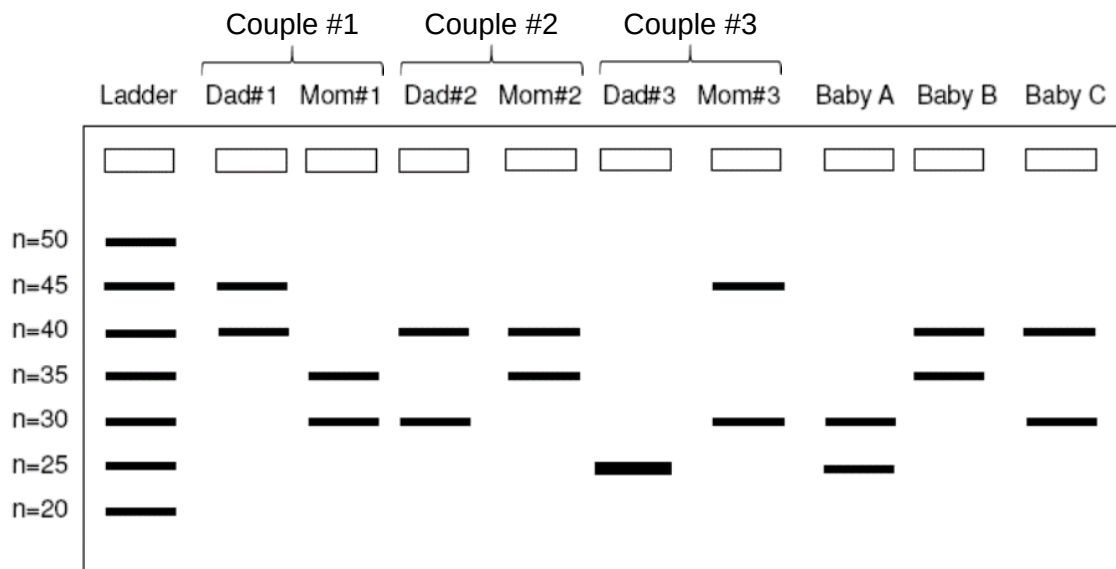


Fig.1.1

- (a)** Describe how the DNA bands in the gel could be made visible.

[1]

- (b)** State the purpose of the DNA ladder.

[1]

(c) Explain how gel electrophoresis is used to separate fragments of DNA.

[illegible]

[5]

(d) Explain the banding pattern of Dad #3.

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[3]

(e) Identify the couple that Baby A belongs to.

[1]

(f) Explain why the results shown in Fig. 1.1 could not confirm which couple that Baby B belongs to.

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[2]

- (g) Suggest an alternative DNA test to identify the couple that each of the three mixed up babies belongs to.

..... [1]

[Total: 14]

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Section B

Answer the question in this section

- 2 Haemophilia is an X-linked recessive disorder that impairs the body's ability to form blood clots. There are two main types, haemophilia A and haemophilia B, resulting from deficiencies in clotting factors VIII and IX respectively.

Both *ex vivo* and *in vivo* gene therapy approaches are undergoing clinical trials.

- (a) (i) Explain why gene therapy approach for treatment of haemophilia is possible.

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 [1]

- (ii) Distinguish between *ex vivo* and *in vivo* gene therapy approaches.

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 [2]

Haematopoietic stem cells (HSC) are ideal vehicles for *ex vivo* gene therapy application.

- (b) (i) Describe the normal function of HSC.

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 [1]

- (ii) Explain why HSC are ideal vehicles for *ex vivo* gene therapy application.

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 [3]

The liver is an ideal target for *in vivo* gene therapy application. As a major organ and central metabolic hub, it receives an abundant blood supply through sinusoids with highly permeable walls, which facilitates easy access of blood-borne particles to the hepatocytes. Hepatocytes are long-lived and robust protein factories that can efficiently release their products into the blood circulation. Both viral and non-viral vectors have been and continued to be investigated.

- (c)** Use of non-viral vectors for gene therapy has met with low success rate.

Describe one method of non-viral vector delivery and explain why the method has achieved low success.

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[3]

- (d)** Many ongoing clinical trials are focused on the application of adeno-associated viral (AAV) vectors, which are found to remain as episomes within the nucleus.

Explain an advantage and a disadvantage of the use of such adeno-associated viral vectors compared to retroviral vectors.

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[2]

- (e) Somatic cell nuclear transfer (SCNT), also known as therapeutic cloning, involves the replacement of an egg nucleus with a somatic cell nucleus. As the oocyte develops into a blastocyst, cells from the inner cell mass can be isolated and purified to serve as a source of pluripotent stem cells.

Explain why such an approach cannot be used to treat patients with haemophilia.

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[2]

[Total: 14]

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Section C

Answer the question in this section.

- 3** The success of culturing callus is low due to the difficult task of removing contaminants. It has proven more challenging for plants taken from the wild in tropical countries. In Sumbawa, Indonesia, plant tissue samples were sampled at three different times of the year. They were grown in medium containing no fungicide or antibiotic.

Table 3.1 shows the results.

Table 3.1

explant	time of year	number of explants	number of cultured explants with no fungal or bacterial contamination	percentage of cultured explants with no fungal or bacterial contamination
leaf disc	April	153	12	8
	August	322	16	5
	January	332	30	9
shoot tip	April	194	116	60
	August	191	122	64
	January	211	156	74

- (a)** Describe and explain the results of using the two types of explant.

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[3]

- (b)** Besides using the appropriate explant, suggest how the number of contaminated samples could be reduced when using wild plant samples.

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[1]

- (c)** Explain, with relevant examples, how genetic engineering has helped to increase the quantity of crops for farmers.

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..... [3]

- (d)** Describe two disadvantages of plant tissue culture.

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..... [2]

- (e)** In several countries including China, Korea and the United States, human DNA has already been put into eggs from both rabbits and cows.

Discuss the ethical concerns of conducting such experiments.

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..... [3]

[Total: 12]

Section D

Answer the question in this section on the answer paper provided.

4 Planning Question

You are required to plan, but not carry out, an investigation into the effect of increasing concentration of glucose on rate of respiration of yeast.

Yeast synthesizes ATP through two major biochemical pathways: aerobic respiration and fermentation. During both aerobic respiration and fermentation, yeast cells break down glucose molecules within the cell to release energy, and some of this energy is captured and stored in the ATP's high-energy phosphate bonds. The breakdown of glucose also releases carbon atoms, which become available for biosynthetic reactions, enabling the yeast to grow and reproduce by budding. The rest of the carbon ends up in the by-products of these reactions.

You are to use methylene blue in your investigation. Methylene blue acts as an artificial electron acceptor during respiration, which changes from blue to colourless as a result of its reduction in the enzymatic reaction.

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure that they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- yeast suspension,
- 10% glucose solution,
- distilled water,
- access to tap water,
- thermostatically controlled water bath,
- methylene blue,
- stopwatch,
- a variety of different sized beakers, test-tubes, boiling tubes, measuring cylinders or syringes for measuring volumes.

[Total: 12]

Section E

Answer the question in this section on the answer paper provided.
Begin each part of the question on a new piece of answer paper.

5 Free-response question

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

(a) Describe the goals, benefits and ethical concerns of human genome project. [8]

(b) Explain the significance of genetic engineering in improving food quality. [6]

(c) Discuss the social and ethical implications of genetically modified crop plants. [6]

[Total: 20]

--- End of Paper ---